Disaster Response

Disaster response in agriculture leverages AI, remote sensing, and sensor networks to **predict**, **monitor**, and **mitigate the impact** of natural disasters such as **floods, droughts, and cyclones**. These tools allow governments, agritech firms, and insurers to deliver **real-time alerts**, assess **damage intensity**, and support **post-disaster recovery** with targeted advisories and insurance calculations.

> # Agriculture loses over **$30 billion annually worldwide** due to climate disasters. Tech-led response systems can reduce impact by up to **40%** with early warning and better preparedness.

Category	Example Product
Predictive Tech	AI/ML-based flood/drought/earthquake forecast system
Real-time Response	Drone + IoT for rescue & supply delivery
Community Alert System	Multilingual voice/SMS alert platform for rural areas
Relief Logistics	Blockchain for transparent aid distribution
Post-Disaster Assessment	Satellite + drone imagery to assess damage
Rural Resilience	WhatsApp-based weather & advisory bot for farmers
Infrastructure Auditing	Mobile app to map unsafe buildings, landslide zones

Indian Startups/Projects in DisasterTech & Their Workflows(Summarized form from the given details)

1. SatSure (Bengaluru) – Satellite Data + AI for Agri & Disasters

Workflow:

1. Collect satellite data (NDVI, rainfall, crop health)
2. AI prediction for droughts, floods, crop losses
3. Supports insurance companies + disaster mitigation + agri advisory
4. Partners with NABARD, HDFC Ergo, govt agencies

Use Case: Helps insurers settle claims post-disaster using satellite proof
USP: Combines geospatial + weather + financial modeling

2. Garuda Aerospace (Chennai) – Drones for Disaster Relief

Workflow:

1. Drones deployed post-floods/cyclones to assess damage
2. Thermal cameras for victim detection
3. Spraying drones for post-disaster sanitation
4. Tied up with NDRF, Tamil Nadu Govt

Use Case: Used in 2023 Tamil Nadu floods for locating trapped families
USP: India's first drone startup with full disaster protocol

SatSure: Detailed Workflow

Data Acquisition & Integration

- Satellite Imagery: Regularly acquires multi-resolution data from optical (e.g., Sentinel-2, Landsat-8) and radar (e.g., Sentinel-1) satellites.
- Weather & Environmental Data: Integrates historical and real-time weather, rainfall, temperature, and soil moisture data.
- Agronomic & Financial Data: Gathers data on crop types, sowing dates, yield history, land ownership, loan history, and market prices.

- On-Ground Validation: Supplements with drone imagery, IoT sensors, and field surveys for model calibration.

Preprocessing & Farm Boundary Extraction

- Farm Parcel Mapping: Uses image segmentation and AI/ML to extract precise field boundaries at scale.
- Cloud-Free Imagery Processing: Applies deep learning fusion of radar and optical images to generate usable, cloud-free composites.
- Edge Computation: Through in-house satellite infrastructure (like KaleidEO), runs certain ML tasks directly on satellites to reduce latency and data overload.

AI/ML Analytics & Risk Index Generation

- Crop Monitoring:
    - Computes indices like NDVI, EVI, SAVI for plant health.
    - Detects sowing, flowering, and harvest stages.
    - Monitors anomalies (e.g., pest stress, drought onset).
- Disaster Impact Analysis:
    - Compares pre- and post-disaster imagery.
    - Detects flooded fields, drought zones, and storm-impacted areas.
    - Estimates percentage crop loss by region or village.
- Financial Risk Modelling:
    - Generates SatScore: A multi-dimensional risk score (crop health, financial history, irrigation access).
    - Supports underwriting and loan eligibility decisions.

Delivery Platforms & Decision Tools

- SatSure Sparta:
    - SaaS dashboard for developers, policy planners, and data scientists.
    - Offers APIs and widgets to plug analytics into existing systems.

- SatSure Sage:
    - End-to-end lending and insurance support platform.
    - Used by NBFCs, microfinance institutions, and banks to assess farm-level creditworthiness, monitor portfolios, and automate crop insurance claim processing.

- Custom Dashboards:
    - Built for government agencies for disaster impact monitoring, war-room dashboards during floods or droughts, and real-time crop condition monitoring.

Use Cases & Stakeholder Support

- Banks & Microfinance:

    - Supports millions of farmers in accessing credit using data-backed risk scores.
    - Monitors default risk using crop conditions during loan tenure.

- Insurance Companies:
    - Reduces claim settlement time by providing crop-loss verification using satellite data.
    - Detects false claims using geotagged damage assessments.
- Government & Disaster Response
    - Provides near real-time damage heatmaps during events like floods or cyclones.
    - Aids in prioritizing relief delivery and rehabilitation efforts.
- Agri Input & Supply Chain Firms: Forecasts demand, procurement planning, and logistics routing based on regional yield outlook.

Innovation & Strategic Vision

- Edge AI Satellites: Developing and launching own constellation with in-built AI processors to accelerate analysis.
- API-first Approach: Platform services can be embedded into third-party apps and CRMs.
- Scale: Covers over 100 million hectares across India, Southeast Asia, and Africa.
- Growth Model: SaaS + Data-as-a-Service (DaaS) + Insights-as-a-Service (IaaS).

This workflow is summarized by articles from Offical page and Story page(For roadmap)

Garuda Aerospace: Detailed Workflow for Disaster Management

Rapid Deployment & Coordination

- Activated upon disaster alerts in coordination with disaster response agencies like municipal authorities or the NDRF.
- Certified pilots and drone units are dispatched to the affected zones.
- Flight paths and mission parameters are pre-mapped or dynamically assigned.

Drone-Based Emergency Supply Delivery

- Delivery drones transport essential supplies such as food packets, water bottles, medicines, and communication tools to remote or inaccessible disaster-hit areas.
- Especially effective during floods, urban blockades, landslides, and cyclone aftermaths.
- Payload drones are capable of navigating unstable terrain where ground vehicles can't reach.

Aerial Surveillance & Real-Time Situational Awareness

- Surveillance drones capture live aerial video and high-resolution imagery of affected zones.

- These are transmitted to command centers for real-time monitoring and situational assessment.
- Helps responders assess water levels, traffic blockages, infrastructure damage, or fire zones.

Damage Assessment & Reconnaissance

- Drones are used to survey and analyze damaged infrastructure, buildings, bridges, farmland, and roads.
- Data is processed into heatmaps or orthomosaics to identify severely impacted areas.
- Supports efficient prioritization of relief delivery and restoration planning.

Search & Rescue Assistance

- Specialized drones with thermal and infrared cameras help detect trapped victims, survivors, or stranded animals.
- Used in tunnel collapses, floods, and post-earthquake situations to guide rescue personnel.
- Reduces response time and improves rescue efficiency in complex terrain or low visibility zones.

Government & Institutional Collaboration

- Operates in partnership with municipal corporations, disaster relief teams, and emergency response forces.
- Local authorities define mission goals while Garuda handles drone logistics, pilots, and data relay.
- Integrated into official disaster mitigation protocols in multiple Indian states.

Technology & Operations Backbone

- Maintains a drone fleet of multiple types: surveillance, delivery, mapping, thermal imaging.
- Operates with hundreds of DGCA-certified pilots across major cities and rural hubs.
- Scales on a Drone-as-a-Service (DaaS) model, enabling rapid multi-location deployment.

3. Ambee (Bengaluru) – Environmental Intelligence

.
.
..

.
..
.

.
..
.
.
.
.
.
.

..
..

Key workflow in DisasterManagent(in tech)

1 Prediction & Detection

Goal: Anticipate disasters (floods, earthquakes, droughts, cyclones) before they occur to reduce impact.

Key Steps:

- **Data Collection:**
  - Aggregate multi-source data: rainfall, river flow, wind speed, terrain elevation, soil moisture, historical disaster events, land use patterns.
  - Incorporate satellite imagery (e.g., Sentinel, MODIS), local weather stations, and hydrology models.

- **Model Development:**
  - Build predictive models using machine learning (e.g., XGBoost, LSTM, ARIMA) and geospatial techniques.
  - Calibrate for specific disasters: flood-prone basins, seismic hotspots, drought zones.

- **Hotspot Forecasting:**
  - Predict potential impact zones 24–72 hours in advance.

    - Generate real-time risk heatmaps and early anomaly flags.

2 Real-time Early Warning

Goal: Communicate targeted, timely, and actionable alerts to at-risk populations.

Key Steps:

- **Alert Dissemination:**
    - Send alerts via SMS, WhatsApp, IVR calls, app notifications in local languages.
    - Integrate alerts with government channels: local Panchayat portals, district emergency centers, TV/radio broadcasts, and public sirens.
- **Localization:** Use geo-targeting to customize alerts by region, severity level, and type of hazard.
- **Automated Escalation:**
    - Prioritize alerts (e.g., “evacuate immediately” vs “watch and prepare”).
    - Integrate dynamic alert thresholds using feedback from IoT sensors and weather APIs.

3 Rescue & Response Coordination

Goal: Enable fast, tech-assisted relief operations and search/rescue efficiency.

Key Steps:

- **Drone Surveillance:**
    - Deploy drones for live aerial imaging, victim location, and damage scanning.
    - Feed video streams to command centers and first responders.
- **Geolocation Tools:**
    - Use GPS and mobile tracking to tag stranded individuals, blocked roads, or dangerous zones.

    - Integrate with rescue mobile apps or chatbots for SOS signaling.

- **IoT Sensor Network:**
    - Install water-level sensors, seismic monitors, or air quality sensors.
    - Use low-power protocols (e.g., LoRaWAN, NB-IoT) to relay real-time data even in low-connectivity zones.

4. Relief Logistics & Aid Distribution

Goal: Ensure fast, fair, and transparent distribution of relief materials and funds.

Key Steps:

- **Central Dashboard:** Create a real-time dashboard for NGOs, governments, and donors to track demand and supply across shelters and districts.

- **Inventory & Tracking:**

    - Use QR/barcode systems on food kits, medicines, tents, and other resources.
    - Monitor aid flow to avoid duplication, delay, or theft.

- **Financial Transparency:** Implement blockchain-based fund tracking to ensure money flows securely from donors to beneficiaries (especially in multi-village operations).

- **Route Optimization:** Use mapping algorithms to optimize delivery routes for trucks, drones, or boats based on blocked roads and access zones.

5 Damage Assessment & Recovery Planning

Goal: Measure impact, trigger insurance or compensation processes, and plan rebuilding.

Key Steps:

- **Image Analysis:**
    - Collect post-disaster imagery via drones or satellites.

    - Use computer vision (CNNs) to classify damaged structures, water-logged fields, or landslides into severity levels (e.g., light, moderate, total loss).

- **Insurance & Relief Automation:**
    - Integrate assessment data with government or insurance portals for quick claim processing.
    - Generate auto-reports for compensation approvals, land-use changes, or restoration needs.
- **Policy & Learning Loop:**
    - Feed collected data back into predictive models to improve future preparedness.
    - Support urban planning and zoning reform based on impact analysis.

How It's Used Abroad :-

Disaster-tech is now widely deployed in:

- **Japan**: Automated flood gates & prediction via AI.
- **Netherlands**: Coastal flooding simulations for urban-agri zones.
- **USA** (NOAA + IBM Weather): Integrated flood/drought forecasting + advisory.
- **Africa** (FAO + Digital Earth Africa): Crop-loss mapping and early warning for food security.

Why India need it :-

High Vulnerability to Climate Shocks

- Over 60% of Indian farmland is rain-fed, making it extremely vulnerable to droughts, erratic rainfall, and heatwaves.
- Climate change is intensifying the frequency and severity of extreme weather events, directly threatening crop yields.

Insurance Burden Driven by Disasters

- Nearly 80% of agricultural insurance claims arise from extreme weather events (floods, droughts, cyclones).

- Faster and more accurate damage assessment through technology can improve compensation timelines and reduce fraud.

Information Deserts in Rural India

- Millions of smallholder farmers lack access to real-time, hyper-local advisory systems.
- Without timely alerts, they miss opportunities to prepare, adapt, or mitigate losses.

Fragmented Traditional Response

- Current disaster response mechanisms are often slow, manual, and disconnected across states and agencies.
- Tech-driven micro-advisories powered by AI, IoT, and mobile platforms can provide personalized, location-specific guidance — revolutionizing how India handles agricultural crises.

Market Size

Global Disaster Management Market

- Global market (all disasters): Valued at approximately USD 95.5 billion in 2024, projected to rise to about USD 162.7 billion by 2033, at a CAGR of ~6.1% from 2025–2033. Consegic Business Intelligence and PreventionWeb Business Research Insights.

- Natural disaster management subset: Expected to grow from USD 63.9 billion in 2024 to USD 114.0 billion by 2029, at a CAGR of ~12.3% Newswire

- Emergency & disaster response: Forecasted to increase from USD 164.7 billion in 2023 to USD 322.7 billion by 2033, translating to a 6.96% CAGR Spherical Insights.

- Natural disaster management (broader): Estimated to reach USD 196.8 billion by 2032, with a 12.5% CAGR between 2025–2032 Acumen Research and Consulting+11Consegic Business Intelligence+11PR Newswire+11.

India Crisis & Disaster Management Market

- Crisis management (India): Grew from USD 6.88 billion in 2024 to a projected USD 17.90 billion by 2032, at a CAGR of ~12.7% Markets and Data.

- Asia-Pacific region (including India): Leads disaster tech adoption; accounts for major share due to frequent natural hazards and strong government initiatives